LOGISIM

CSE 206

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What is Logisim?

- Tool for designing and simulating digital logic circuits
- Developed by Dr. Carl Burch

Download EXE or JAR

- Make sure JDK/JRE is installed
- Download EXE (Requires JRE 1.5.0)
- https://sourceforge.net/projects/circuit/files/2.7.x/2.7.1/

OR

- Download Jar (Works for JRE >= 1.5.0)
- https://sourceforge.net/projects/circuit/files/2.7.x/2.7.1/logisim-generic-2.7.1.jar/download
- Double click the exe/jar to open Logisim

Resources

- Follow demonstration
- Useful Resources
- http://www.cburch.com/logisim/docs/2.7/en/html/guide/index.html
- https://www.youtube.com/watch?v=cMz7wyY_PxE&ab_channel=BarryBrown

Implement a Basic Circuit

- $(\bar{A} \cup B) \cap (\overline{c \cap A})$

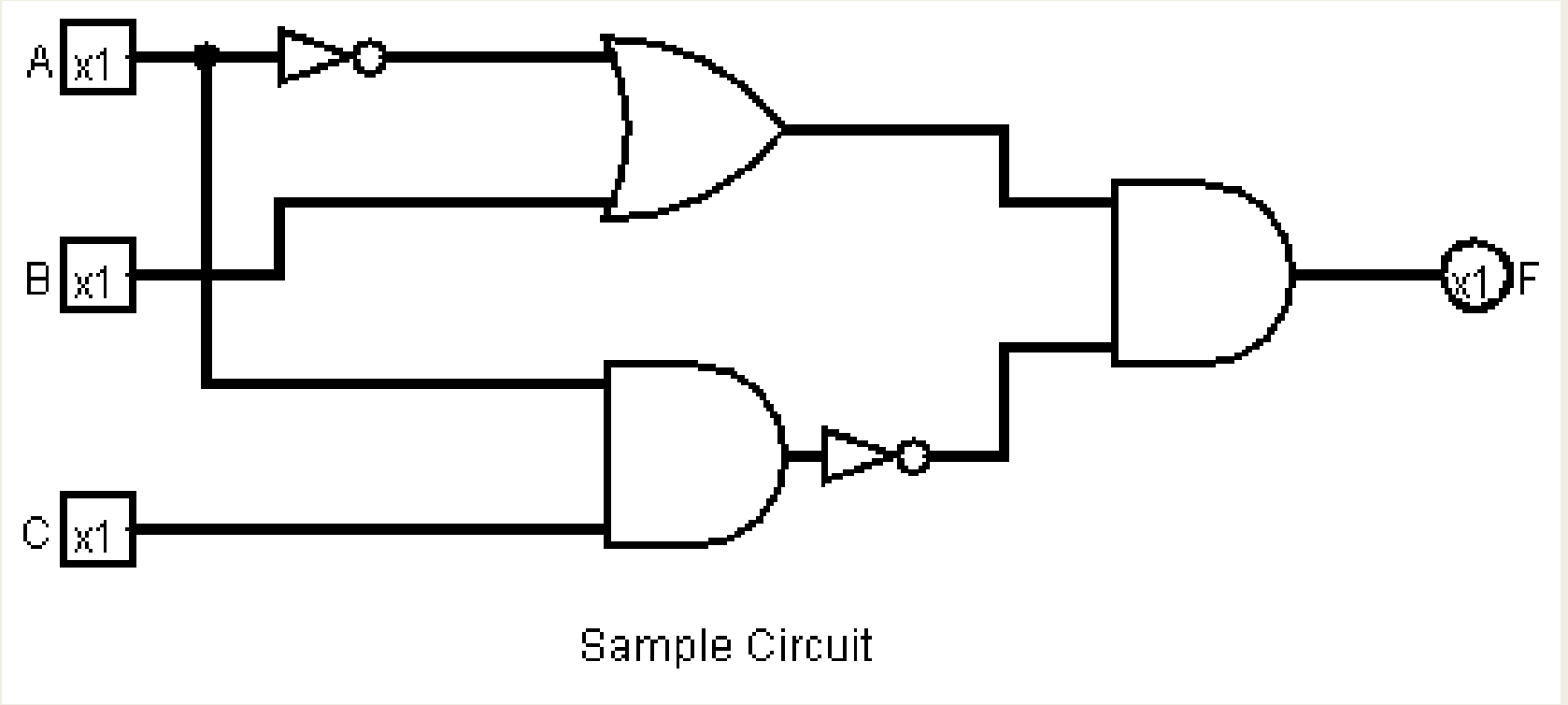
Build the Truth Table

- $(\bar{A} \cup B) \cap (\overline{c \cap A})$

A	B	C	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Design in Logisim and Simulate

- $(\bar{A} \cup B) \cap (\overline{c \cap A})$
- Match output with the truth table



Sample Circuit

View Truth Table in Logisim

- $(\bar{A} \cup B) \cap (\overline{c \cap A})$
- Project -> Analyze Circuit

A	B	C	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Practice

- Implement XOR gate with AND, OR and NOT gates
- Implement Boolean functions with Basic Gates
- $F(A, B, C, D) = A + \overline{BC} + D\bar{A} + \overline{(C + D)}$
- Draw the truth table on notebook and match the results with the simulator